(Please write your Exam Roll No.) *Exam Roll No. .....*

END TERM EXAMINATION

SIXTH SEMESTER [B.TECH.] JULY 2023

Paper Code: ETEC-310 ***Subject: Data communication and Networks***

(IT/ECE/MAE/ITE)

Time: 3 Hours ***Maximum Marks: 75***

Note: Attempt any five questions including Q.No1 which is compulsory. Select one question from each unit.

Q1 Attempt **any Five** **[5x5=25]**

(a) Explain the correlation between the OSI model and TCP/IP model with a proper diagrammatical form.

(b) What do you understand by Circuit Switching and Packet Switching? Discuss through the diagram.

(c) Describe the LAN-based FDDI over a basic token ring in detail.

(d) Differentiate SMTP & POP protocol.

(e) Explain the IPv4 protocol with header format. Also, compare with IPv6.

(f) Explain the High-Level Data Link Control (HDLC) protocol used in computer networks. Discuss its key features, frame structure, and modes of operation.

UNIT-I

Q2 a) What is the OSI reference model? Draw a block diagram and explain the functioning of each layer. **[7.5]**

b) i. Define the transmission modes with a diagram. **[2.5]**

ii. Describe the transmission impairment. **[2.5]**

Q3 Explain the following terms with proper examples: **(2.5x5=12.5)**

a) Virtual Circuit with diagram

b) Guided and Unguided Media

c) Wavelength

d) Differentiate Process to Process and Host to Host delivery

e) Baseband transmission and Broadband transmission

UNIT-II

Q4 a) Explain the working of the Go-Back-N ARQ protocol and the Selective Repeat ARQ protocol. Also, compare the Sender and Receiver Window Sizes of both ARQ. **[6.5]**

b) Short note on: **[6]**

i) Leaky bucket algorithm

ii) Channelization in Medium Access Control

Q5 a) Define the meaning of carrier sense in CSMA. Also, Explain the CSMA/CD and CSMA/CA with the diagram. **[6.5]**

b) Explain the following: **[6]**

i. Control Protocols in Medium Access Control

ii. Pure ALOHA and Slotted ALOHA

P.T.O.